

Submitted by	_____	Course	_____
Roll number	_____	Instructor	_____
Department	_____	Date	_____

1 PROJECT TITLE

Nuskha — Prescription and Discharge Instruction Agent. An AI agent that reads a prescription or hospital discharge slip and converts it into a daily medicine schedule and plain-language instructions that the patient or their family can actually follow at home.

2 PROBLEM STATEMENT

A patient leaves the hospital holding a slip that was written for another doctor to read. It says things like **1 BD x 7d, ac** in handwriting, lists four or five medicines by brand name, and adds a review date somewhere at the bottom.

The explanation that should go with it usually does not happen. Discharge takes a few minutes on a busy ward, the pharmacy counter is crowded, and the person who ends up giving the medicines at home is often a son or daughter who was not in the room. Elderly patients and patients who do not read English are worst affected.

The failures that follow are predictable. Doses get missed or doubled. Two different brand names containing the same salt are taken together. Antibiotics are stopped as soon as the fever drops. Instructions such as "empty stomach" or "avoid milk" are ignored because nobody explained why they were written. Worst of all, the warning signs that should send the patient straight back to hospital sit unread on a folded slip in a drawer.

3 PROPOSED SOLUTION

Nuskha takes a photo of the slip and produces four things: a schedule with real clock times, a one-line explanation of each medicine in Urdu or English, the food and timing rules that apply, and the list of symptoms that mean go back to hospital. Follow-up questions can then be asked in conversation.

The design rule behind the whole project is that the agent explains what the doctor wrote and nothing more. It does not diagnose, does not suggest medicines, and does not change a dose. Where it cannot read a field with confidence, it says so and asks the user to confirm with the pharmacist instead of filling the gap with a guess. Safety checks such as duplicate salts or known interactions are shown as questions to raise with the doctor, never as an instruction to stop or alter anything.

4 TARGET USERS

- Patients discharged after surgery, admission or a new chronic diagnosis, who go home with several medicines at once.
- Family caregivers who manage medicines for a parent, spouse or child.
- Elderly patients and patients with limited English or limited literacy.
- Rural patients whose nearest pharmacist is not available to re-explain the slip.
- Pharmacists and nursing staff, as a quick counselling aid at the counter (secondary user).

5 OBJECTIVES

- Turn a prescription or discharge slip into an unambiguous, day-by-day schedule.
- Expand medical shorthand and abbreviations into everyday Urdu or English.

- Detect duplicate salts, allergy conflicts and interactions among the listed medicines, and surface them as questions for the doctor.
- Pull the follow-up date and the return-to-hospital warning signs out of the discharge text and put them in front of the patient.
- Keep every line of output traceable to a line on the slip.
- Refuse to guess. Low-confidence fields are reported as unread, not invented.

6 INPUT

Input	Form	Notes
Prescription	Photo or scan	Handwritten or printed, single or multiple pages.
Discharge slip	Photo or PDF	Contains the follow-up date and warning signs.
Patient profile	Short form	Age, known allergies, medicines already being taken, language preference, usual meal times.
Questions	Text or voice	Asked by the patient or caregiver after the schedule is generated.

7 AI AGENT PROCESSING / WORKFLOW

Stages 1 to 7 run once per uploaded slip. The question-answering loop then runs as often as the user needs it, using the structured data produced by the first pass.

1. Image preprocessing and OCR. The photo is deskewed, cropped and contrast-corrected, then passed through OCR. Handwriting and printed text are handled by different models and the better result is kept.

2. Field extraction. Each prescription line is broken into medicine name, strength, dosage form, frequency code, route, duration and any special instruction. The output is a structured record, not free text.

3. Medicine identification. Brand names are matched against a drug database and mapped to their generic salt. Close misspellings from OCR are resolved by fuzzy matching against the same list.

4. Confidence check. Every extracted field carries a confidence score. Anything below the threshold is separated out and reported to the user as unclear. This gate is what stops a misread strength from reaching the schedule.

5. Safety checks. The agent looks for two medicines carrying the same salt, known interactions inside the prescribed set, and conflicts with the allergies in the patient profile. Findings are written as questions for the doctor.

6. Schedule construction. Frequency codes and food rules are converted into actual clock times using the patient's stated wake-up and meal times, and expanded across the full course length.

7. Plain-language generation. Each medicine gets one line saying what it is broadly for and how to take it, written in the chosen language. The warning signs and follow-up date are extracted and placed at the top of the output, where they will be seen.

8. Question answering. Later questions are answered only from the structured record built above. If the slip does not contain the answer, the agent says so and points the user to the pharmacist.

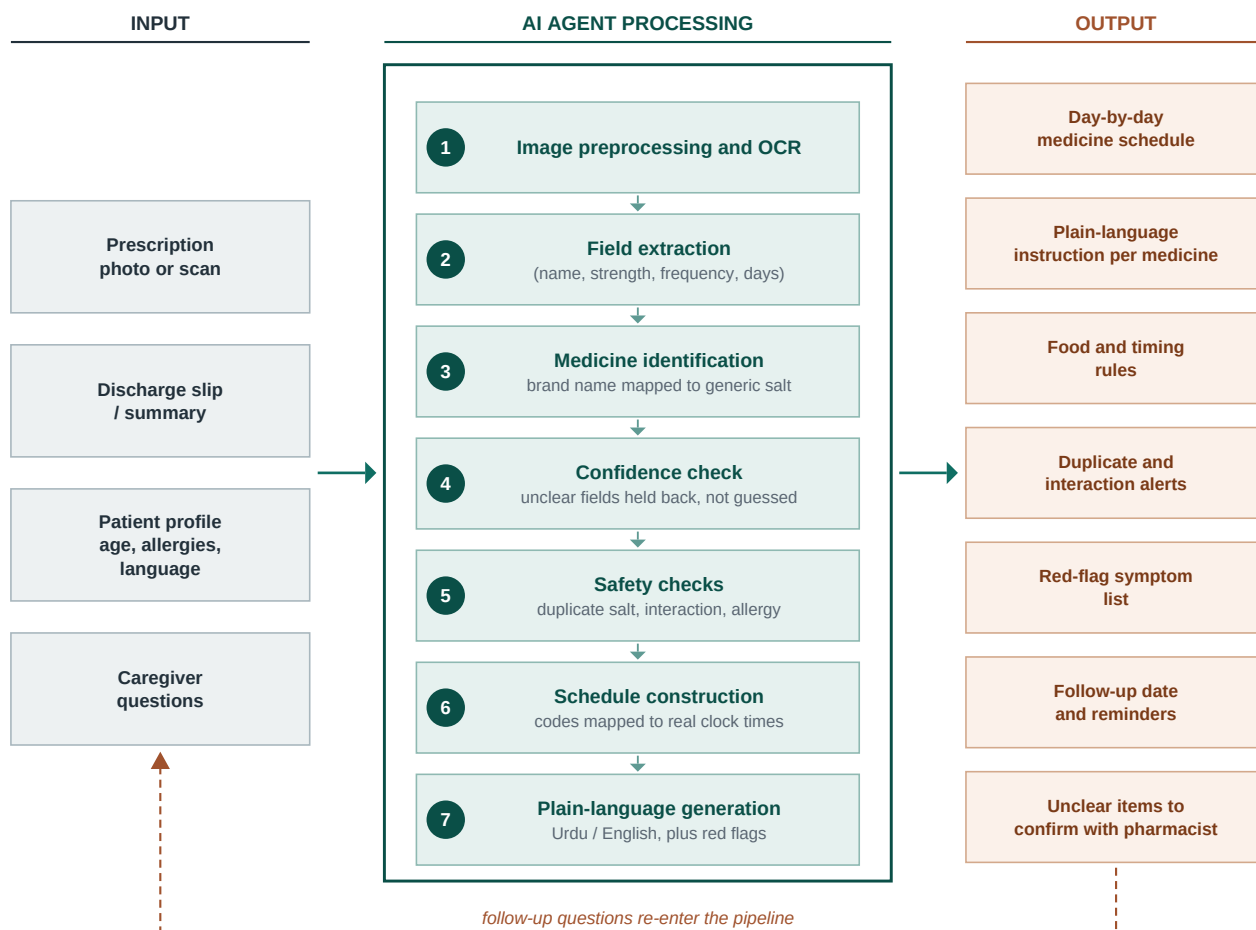


Figure 1. Input, processing and output structure of the agent. Stage 4 is the safety gate: fields that cannot be read confidently leave the pipeline as items to confirm rather than continuing into the schedule.

8 OUTPUT

- A **day-by-day schedule** with clock times for each medicine and the date the course ends.
- A **one-line explanation** for every medicine, in Urdu or English.
- **Food and timing rules**: before or after meals, with water, avoid milk, and so on.
- **Duplicate and interaction alerts**, phrased as points to raise with the doctor.
- A **red-flag list**: the symptoms that mean return to hospital immediately.
- The **follow-up date**, what to bring to it, and a reminder set for it.
- A list of **fields the agent could not read**, for confirmation at the pharmacy.
- **Answers** to caregiver questions, based only on the parsed slip.

9 KEY FEATURES

Feature	Description
Photo input	Works from a phone photo of a handwritten or printed slip.
Abbreviation expansion	BD, TDS, OD, SOS, ac, pc, HS and similar codes are translated.
Brand to salt mapping	Identifies the actual drug behind the brand name.
Confidence gate	Unreadable fields are reported, never guessed.
Duplicate salt alert	Catches the same ingredient arriving under two names.

Feature	Description
Interaction check	Limited to the medicines on the slip and the patient profile.
Allergy conflict check	Cross-checks against allergies entered by the user.
Time-based schedule	Converts frequency codes into a real daily timetable.
Bilingual output	Urdu or English, chosen by the user.
Red-flag extraction	Lifts warning signs out of the discharge text.
Follow-up reminder	Appointment date captured and scheduled.
Caregiver Q&A	Conversational follow-up grounded in the parsed slip.

10 EXAMPLE USER INTERACTION

A patient is discharged after a minor wound infection and uploads the slip. The left column is what is written on it; the right column is what the agent shows the patient.

On the slip	What the patient sees
T. Augmentin 625mg 1 BD x 7d, pc	Augmentin 625 mg tablet — antibiotic. One tablet twice a day, at 8:00 am and 8:00 pm, just after food, for 7 days (until 20 September). Finish all 14 tablets even if you feel better.
Cap. Omeprazole 20mg 1 OD, ac	Omeprazole 20 mg capsule — protects the stomach. One capsule daily, 30 minutes before breakfast, at 7:30 am.
T. Panadol 500mg 1 SOS, max 4/day	Panadol 500 mg — for pain or fever, only when needed. Never more than 4 tablets in 24 hours. <i>Note: this contains paracetamol. Check any other fever medicine at home for the same ingredient so you do not take a double dose.</i>
Review OPD after 10d. Return if fever >101°F, vomiting, wound discharge.	Go back to hospital immediately if: fever above 101°F, repeated vomiting, or any discharge from the wound. Follow-up: OPD on 23 September. Bring this slip and all medicine boxes.

Held back for confirmation: the strength on line 4 could not be read with confidence. The agent shows the line as written and asks the user to confirm it with the pharmacist before starting that medicine.

Caregiver: "Can I give both the 8 pm tablets together?"

Agent: "Only Augmentin is due at 8:00 pm. Omeprazole is a morning medicine and should stay before breakfast. Panadol is not on the schedule, it is only for pain or fever."

Caregiver: "He vomited the 8 am dose. Should I repeat it?"

Agent: "The slip does not cover this and it is not something I can decide. Call the prescribing doctor or your pharmacist before giving another dose."

11 BENEFITS

- The instruction reaches the person who actually gives the medicine at home, not just the person who was standing at the counter.
- Abbreviations stop being a barrier, which removes a common cause of wrong dosing.
- Duplicate-salt and interaction warnings catch a class of mistake that patients cannot catch themselves.
- Warning signs and the follow-up date are pushed to the top instead of being buried in the slip.
- It costs the hospital nothing in staff time, which matters in an OPD seeing hundreds of patients a day.

- Output in Urdu makes the instructions usable by patients who would otherwise depend on someone else to read for them.

12 LIMITATIONS

- Handwriting OCR is the weakest part of the system. Poor handwriting or a bad photo will produce unread fields, which is by design safer than a wrong reading but is still a failure.
- Coverage depends on the drug database. Local brands that are missing from it cannot be mapped to a salt, so no safety check runs on them.
- Interaction checking is limited to the medicines the agent knows about. It cannot see anything the patient is taking but did not declare.
- The agent cannot catch a prescribing error. If the doctor wrote the wrong dose, the agent will faithfully schedule the wrong dose.
- Urdu medical vocabulary is inconsistent and often mixed with English, so translation quality varies and needs review by a clinician.
- It is a communication tool, not a clinical one. Real deployment would need hospital approval, a pharmacist in the loop and clear legal wording about responsibility.

13 FUTURE IMPROVEMENTS

- Direct integration with hospital systems, so the prescription arrives as structured data and OCR is skipped entirely.
- Voice output and voice questions, for patients who cannot read either language.
- Photo-based verification: the patient photographs the medicines bought, and the agent checks them against the prescription.
- Dose reminders with a simple confirmation tap, and a missed-dose log the doctor can see at the follow-up visit.
- Caregiver mode for managing more than one patient, which is common in joint families.
- A pharmacist review step, where a licensed pharmacist can approve or correct the generated plan before it reaches the patient.
- Refill tracking, so a chronic patient is warned before the medicine runs out.